

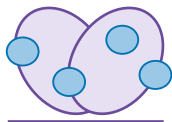
# Prior literature, observed pathway shifts, and hypotheses to test

## Established phenotype from prior literature

## Observed lower expiMap scores in flight

## Complementary tissue-state hypothesis to test

### A THYMUS



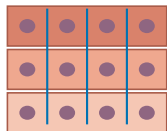
Involution with reduced thymocyte proliferation and adaptive activity [6-9]

- ↓ DNA repair
- ↓ RHOA cytoskeletal cycle
- ↓ Lymphoid-stromal interactions



Known involution may also involve lower repair, thymocyte motility, and stromal-niche coordination.

### B SKIN



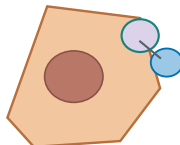
Barrier injury, inflammation, and impaired regeneration [12-16]

- ↓ Chromatin-modifying enzymes
- ↓ DNA repair
- ↓ Hedgehog signaling
- ↓ Sphingolipid metabolism
- ↓ Cell-cell junction organization



Barrier injury may include a broader maintenance deficit spanning regulation, repair, barrier lipids, and cell coordination.

### C LIVER



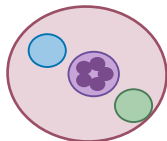
Metabolic and xenobiotic dysregulation [17-23]

- ↓ MHC class II antigen presentation
- ↓ T-cell receptor signaling



Metabolic heterogeneity may coexist with lower adaptive immune communication.

### D SPLEEN



Lower T-cell abundance, activation, and responsiveness [35-37]

- ↓ T-cell receptor signaling
- ↓ Neutrophil degranulation
- ↓ C-type lectin receptor signaling



Reduced adaptive activity may extend to lower innate sensing and effector-related transcription.

Prior literature, this study's observations, and new hypotheses are shown separately; dotted arrows denote inference, not causality. Lower scores do not prove pathway inhibition, and cell composition may contribute.